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MATHEMATICS - 2008

Objective Questions: (Q.1-Q.15)

- (1.) The least positive integer n , such that

$$\begin{bmatrix} \cos \frac{\pi}{4} & \sin \frac{\pi}{4} \\ -\sin \frac{\pi}{4} & \cos \frac{\pi}{4} \end{bmatrix}^n$$
 is the identity matrix of order 2,

is

- (a.) 4
- (b.) 8
- (c.) 12
- (d.) 16

- (2.) Let $S = \{T : \mathbf{R}^3 \rightarrow \mathbf{R}^3 \mid T \text{ is a linear transformation with } T(1, 0, 1) = (1, 2, 3), T(1, 2, 3) = (1, 0, 1)\}$.

Then S is

- (a.) A singleton set
- (b.) A finite set containing more than one element
- (c.) A countably infinite set
- (d.) An uncountable set

- (3.) Let $S_n = \int_0^1 \frac{nx^{n-1}}{(1+x)} dx$ for $n \geq 1$. Then as $n \rightarrow \infty$, the sequence $\{S_n\}$ tends to
- (a.) 0
 - (b.) $\frac{1}{2}$
 - (c.) 1
 - (d.) $+\infty$
- (4.) The work done by the force $\vec{F} = 4y \hat{i} - 3xy \hat{j} + z^2 \hat{k}$ in moving a particle over the circular path $x^2 + y^2 = 1, z = 0$ from $(1, 0, 0)$ to $(0, 1, 0)$ is
- (a.) $\pi + 1$
 - (b.) $\pi - 1$
 - (c.) $-\pi + 1$
 - (d.) $-\pi - 1$
- (5.) The set of all boundary point of $\mathbf{Q}$ in $\mathbf{R}$ is
- (a.) $\mathbf{R}$
 - (b.) $\mathbf{R} \setminus \mathbf{Q}$
 - (c.) $\mathbf{Q}$
 - (d.) ϕ
- (6.) Let $V = \left\{ (x, y, z) \in \mathbf{R}^3 : \frac{1}{4} \leq x^2 + y^2 + z^2 \leq 1 \right\}$

and $\vec{F} = \frac{x \hat{i} + y \hat{j} + z \hat{k}}{(x^2 + y^2 + z^2)^2}$ for $(x, y, z) \in V$.

Let $\hat{n}$ denote the outward unit normal vector to the boundary of V and S denote the part

$$\left\{ (x, y, z) \in \mathbf{R}^3 : x^2 + y^2 + z^2 = \frac{1}{4} \right\}$$

of the boundary of V .

Then $\iint_S \vec{F} \cdot \hat{n} \, dS$ is equal to

- (a.) -8π
- (b.) -4π
- (c.) 4π
- (d.) 8π

(7.) The set $U = \left\{ x \in \mathbf{R} \mid \sin x = \frac{1}{2} \right\}$ is

- (a.) Open
- (b.) Closed
- (c.) Both open and closed
- (d.) Neither open nor closed

(8.) Let $f(x) = \int_0^x (x^2 - t^2) g(t) \, dt$, where g is a real valued continuous function on $\mathbf{R}$. Then $f'(x)$ is equal to

- (a.) 0
- (b.) $x^3 g(x)$

$$(c.) \int_0^x g(t) dt$$

$$(d.) 2x \int_0^x g(t) dt$$

- (9.) Let $y_1(x)$ and $y_2(x)$ be linearly independent solutions of the differential equation $y'' + P(x)y' + Q(x)y = 0$, where $P(x)$ and $Q(x)$ are continuous functions on an interval I . Then $y_3(x) = ay_1(x) + by_2(x)$ and $y_4(x) = cy_1(x) + dy_2(x)$ are linearly independent solutions of the given differential equation if

- (a.) $ad = bc$
- (b.) $ac = bd$
- (c.) $ad \neq bc$
- (d.) $ac \neq bd$

- (10.) The set $R = \{f \mid f \text{ is a function from } \mathbf{Z} \text{ to } \mathbf{R}\}$ under the binary operation '+' and " defined as $(f+g)(n) = f(n) + g(n)$ and $(f \cdot g)(n) = f(n)g(n)$ for all $n \in \mathbf{Z}$ forms a ring.

Let $S_1 = \{f \in R \mid f(-n) = f(n) \text{ for all } n \in \mathbf{Z}\}$

and $S_2 = \{f \in R \mid f(0) = 0\}$.

Then

- (a.) S_1 and S_2 are both ideals in R

- (b.) S_1 is an ideal in R while S_2 is not
- (c.) S_2 is an ideal in R while S_1 is not
- (d.) Neither S_1 nor S_2 is an ideal in R

(11.) Let $T : \mathbf{R}^3 \rightarrow \mathbf{R}^3$ be a linear transformation such that $T(1, 2, 3) = (1, 2, 3)$, $T(1, 5, 0) = (2, 10, 0)$ and $T(-1, 2, -1) = (-3, 6, -3)$. The dimension of the vector space spanned by all the eigenvectors of T is

- (a.) 0
- (b.) 1
- (c.) 2
- (d.) 3

(12.) Let $\{a_n\}$ and $\{b_n\}$ be sequences of real numbers defined as $a_1 = 1$ and for $n \geq 1$,

$$a_{n+1} = a_n + (-1)^n 2^{-n}, \quad b_n = \frac{2a_{n+1} - a_n}{a_n}.$$

Then

- (a.) $\{a_n\}$ converges to zero and $\{b_n\}$ is a Cauchy sequence
- (b.) $\{a_n\}$ converges to a non-zero number and $\{b_n\}$ is a Cauchy sequence
- (c.) $\{a_n\}$ converges to zero and $\{b_n\}$ is not a convergent sequence

(d.) $\{a_n\}$ converges to a non-zero number and $\{b_n\}$ is not a convergent sequence

(13.) Let $f : (-1, 1) \rightarrow \mathbf{R}$ be defined as $f(x) = \frac{x^2}{1 - \cos x}$

for $x \neq 0$ and $f(0) = 2$. If $f(x) = \sum_{n=0}^{\infty} a_n x^n$ is the Taylor expansion of f for all x in $(-1, 1)$, then

$$\sum_{n=0}^{\infty} a_{2n+1} \text{ is}$$

(a.) 0

(b.) $\frac{1}{2}$

(c.) 1

(d.) 2

(14.) Let $y_1(x)$ and $y_2(x)$ be twice differentiable functions on an interval I satisfying the

differential equations $\frac{dy_1}{dx} - y_1 - y_2 = e^x$ and

$2\frac{dy_1}{dx} + \frac{dy_2}{dx} - 6y_1 = 0$. Then $y_1(x)$ is

(a.) $C_1 e^{-2x} + C_2 e^{3x} - \frac{1}{4} e^x$

(b.) $C_1 e^{2x} + C_2 e^{-3x} + \frac{1}{4} e^x$

(c.) $C_1 e^{2x} + C_2 e^{-3x} - \frac{1}{4} e^x$

$$(d.) C_1 e^{-2x} + C_2 e^{3x} + \frac{1}{4} e^x$$

(15.) Let G be a finite group and H be a normal subgroup of G of order 2. Then the order of the centre of G is

- (a.) 0
- (b.) 1
- (c.) An even integer ≥ 2
- (d.) An odd integer ≥ 3

Subjective Questions: (Q.16-Q.29)

(16.) (a.) Let f and g be continuous functions on $\mathbf{R}$

such that $f(x) = \int_0^x g(t) dt$ and $g(x) = \int_x^0 f(t) dt + 1$.

Prove that $(f(x))^2 + (g(x))^2 = 1$ for all $x \in \mathbf{R}$.

(b.) Let $f: \mathbf{R} \rightarrow \mathbf{R}$ be a function such that f' is continuous on $\mathbf{R}$.

Show that the series $\sum_{n=1}^x \left[f\left(\frac{x}{2n}\right) - f\left(\frac{x}{2n+1}\right) \right]$

converges uniformly on $[0, 1]$.

(17.) (a) Find the maxima, minima and saddle points, if any, for the function

$$f(x, y) = (y - x^2)(y - 2x^2) \text{ on } \mathbf{R}^2.$$

(b.) Let $P(x) = a_0 + a_1x^2 + a_2x^4 + \dots + a_nx^{2n}$, where $n \geq 1$ and $a_k > 0$ for $k = 0, 1, \dots, n$. Show that $P(x) - xP'(x) = 0$ has exactly two real roots.

(18.) (a.) Given that $y_1(x) = x$ is a solution of $(1+x^2)y'' - 2xy' + 2y = 0$, $x > 0$. Find a second linearly independent solution.

(b.) Solve: $x^2y'' + xy' - y = 4x \log x$, $x > 0$.

(19.) (a.) Let ϕ be a differentiable function on $[0, 1]$ satisfying $\phi'(x) \leq 1 + 3\phi(x)$ for all $x \in [0, 1]$ with $\phi(0) = 0$. Show that $3\phi(x) \leq e^{3x} - 1$.

(b.) If $y_1(x) = x(1-2x)$, $y_2(x) = 2x(1-x)$ and $y_3(x) = x(e^x - 2x)$ are three solutions of a non-homogenous linear differential equation $y'' + P(x)y' + Q(x)y = R(x)$, where $P(x)$, $Q(x)$ and $R(x)$ are continuous functions on $[a, b]$ with $a > 0$, then find its general solution.

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(20.) (a.) Evaluate: $\int_1^4 \int_0^1 \int_{2y}^2 \frac{\cos x^2}{\sqrt{z}} dx dy dz$

(b.) Find the surface area of the portion of the cone $z^2 = x^2 + y^2$ that is inside of cylinder $z^2 = 2y$.

(21.) (a.) Use Green's theorem to evaluate the integral $\oint_C x^2 dx + (x + y^2) dy$, where C is the closed curve given by $y = 0$, $y = x$ and $y^2 = 2 - x$ in the first quadrant, oriented counter clockwise.

(b.) Let $f : \mathbf{R} \rightarrow \mathbf{R}$ be a continuous function.

Use change of variables to prove that

$$\iint_D f(x-y) dx dy = \int_{-1}^1 f(u) du$$

where $D = \{(x, y) \in \mathbf{R}^2 : |x| + |y| \leq 1\}$.

(22.) Using Gauss's divergence theorem, evaluate the integral $\iint_S \vec{F} \cdot \hat{n} dS$, where $\vec{F} = 4xz \hat{i} - y^2 \hat{j} + 4yz \hat{k}$,

S , is the surface of the solid bounded by the sphere $x^2 + y^2 + z^2 = 10$ and the paraboloid $x^2 + y^2 = z - 2$, and $\hat{n}$ is the outward unit normal vector to S .

(23.) (a.) A square matrix M of order n with complex entries is called skew Hermitian if

$M + \bar{M}^T = 0$, where 0 is the zero matrix of order n .

Determine whether $V = \{M \mid M \text{ is a } 2 \times 2 \text{ skew Hermitian matrix}\}$ is a vector space over

- (i) The field $\mathbf{R}$ and
- (ii) The field $\mathbf{C}$ with the usual operations of addition and scalar multiplication for matrices?

(b.) Let $V = \{P(x) \mid P(x) \text{ is a polynomial of degree } \leq n \text{ with real coefficient}\}$ and $T: V \rightarrow \mathbf{R}^m$ be defined as $T(P(x)) = (P(1), P(2), \dots, P(m))$. Then show that T is linear and determine the Nullity of T .

(24.) Let G be the set of all 3×3 real matrices M such that $MM^T = M^T M = I_3$ and let $H = \{M \in G \mid \det M = 1\}$, where I_3 is the identity matrix of order 3. Then show that

- (i) G is a group under matrix multiplication.
- (ii) H is a normal subgroup of G .
- (iii) $\phi: G \rightarrow \{-1, 1\}$ given by $\phi(M) = \det M$ is onto
- (iv) $\frac{G}{H}$ is abelian

(25.) (a.) Suppose that $(R, +, \cdot)$ is a ring having the property $a \cdot b = c \cdot a \Rightarrow b = c$, when $a \neq 0$.

Then prove that $(R, +, \cdot)$ is a commutative ring.

(b.) Let R be a commutative ring with identity.

For $a_1, a_2, \dots, a_n \in R$, the ideal generated by

$\{a_1, a_2, \dots, a_n\}$ is given by $\langle a_1, a_2, \dots, a_n \rangle$

$$= \{r_1 a_1 + r_2 a_2 + \dots + r_n a_n \mid r_i \in R, 1 \leq i \leq n\}$$

Let $\mathbf{Z}[x]$ be the set of all polynomials with integer coefficients. Consider the ideal

$I = \{f \in \mathbf{Z}[x] \mid f(0) \text{ is an even integer}\}$. Prove

that $I = \langle 2, x \rangle$ and that it is a maximal ideal.

(26.) For a given positive integer $n > 1$, show that there exist subspaces $X_1, X_2, \dots, X_n$ of $\mathbf{R}^m$ for some integer $m > n$ and a linear transformation $T : \mathbf{R}^m \rightarrow \mathbf{R}^m$ such that

- $\dim X_k = k, \quad k = 1, 2, \dots, n$
- For $i \neq j, X_i \cap X_j = \{\vec{0}\}$, where $\vec{0}$ is the zero vector of $\mathbf{R}^m$
- $T(X_k) = X_{k-1}, \quad k = 1, 2, \dots, n$, where $X_0 = \{\vec{0}\}$.

Also, find the rank of T .

(27.) Let $f : (0, \infty) \rightarrow (0, \infty)$ be a continuously differentiable function and let $z = \frac{xy}{f(x^2 + y^2)}$ be defined for $xy \neq 0$.

(a.) Prove that
$$\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = \frac{(x+y)}{[f(x^2 + y^2)]^2}$$

$$\{f(x^2 + y^2) - 2xy f'(x^2 + y^2)\}$$

(b.) Further, if f is homogenous of degree $\frac{1}{2}$,

then verify that $x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = z$.

(28.) Determine the interval of convergence of the power series $\sum_{n=1}^{\infty} n(2n-1)x^{2n}$ and show that its sum is $\frac{x^2(1+3x^2)}{(1-x^2)^3}$ at any point x in its interval of convergence.

(29.) (a.) Let $f : \mathbf{R}^2 \rightarrow \mathbf{R}$ be defined as

$$f(x, y) = x^2 \cos\left(\frac{y}{x}\right) \text{ for } x \neq 0 \text{ and}$$

$$f(x, y) = 0 \text{ for } x = 0. \text{ Compute } \frac{\partial f}{\partial x} \text{ at all}$$

points in $\mathbf{R}^2$ and show that it is continuous at the origin.

(b.) Let $f : (0, 1) \rightarrow (0, \infty)$ be a uniformly continuous function. If $\{x_n\}$ is a Cauchy sequence in $(0, 1)$, then prove that $\{f(x_n)\}$ is a Cauchy sequence in $(0, \infty)$. Hence deduce that for any two Cauchy sequences $\{x_n\}$ and $\{y_n\}$ in $(0, 1)$, $\{|f(x_n) - f(y_n)|\}$ is a Cauchy sequence in $(0, \infty)$.



Rajendra Dubey (Known as Dubey Sir): He has experience of almost three decades of teaching mathematics. During his journey, he has guided IIT-JEE, IAS, NET, GATE, IIT-JAM aspirants etc. Apart from producing top rankers in the various exams, he is widely known for making mathematics interesting and fascinating. He is founder & director of DIPS ACADEMY and chairman of Dubey's Group of Companies.



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